

Møller Polarimeter for HAPPEX

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Current status:

- 24 measurements in April-May 1999
- 8 measurements in July 1999
- Recent studies of the systematic errors
 - Target polarization
 - Dead time
 - Cross-calibration with the Mott polarimeter and with the Møller polarimeters of Halls B and C

Systematic Errors

$$\sigma \propto (1 + A_{ZZ}(\cos \theta_{CM}) \cdot \mathcal{P}_{targ} \cdot \mathcal{P}_{beam})$$

$$A_{obser} = \frac{N_{+-} - N_{-+}}{N_{++} + N_{--}} \approx \overline{A_{ZZ}} \mathcal{P}_{targ} \mathcal{P}_{beam} + \Delta_{transv} - \Delta_{BG} - \Delta_{DT}$$

- $\overline{A_{ZZ}} \sim 0.758 - 0.772$ - the average analyzing power, calculated using simulation $\pm 0.3\%$
 - $\mathcal{P}_{targ} = \mathcal{P}_{foil} \cdot \cos \alpha_{targ}$ - target polarization
 - $\mathcal{P}_{foil}$ - magnetization measurement $\pm 2\%$
 - $\cos \alpha_{targ}$ target – beam angle.
Measurements at $\sim 25\&155^\circ$ $\pm 0.5\%$
 - Δ_{transv} influence of transverse polarization, Measuring at $\alpha_{targ} \sim 25\&155^\circ$ helps $\pm 0.2\%$
 - Δ_{BG} non-polarized background, $\sim 30 - 70\%$ in single arm, depending on E_{beam} , low in coincidence. Accidental coincidences are subtracted. $< 1\%$
 - Δ_{DT} dead time of electronics, estimated $20ns \times 1.MHz$ $\Delta_{DT} \sim 2\%$, LED: $\sim 3\%$ $< 2\%$
 - Observed fluctuations, not understood yet 1%
- Systematic error is estimated: 3.2%

Target Polarization

Ferromagnetic foil, magnetized in a magnetic field.

Foil : supermendur $13.7\mu\text{m}$

Field: $200\text{-}800\text{ Gs}$

$$\mathcal{P}_{foil} = B_{foil} \cdot \frac{g'-1}{2\pi g'} \cdot \frac{1}{N_e \mu_B}, \text{ where}$$

B_{foil} - magnetic field in the foil

$g' = 1.900 \pm 0.005$ - spin/orbital correction for supermendur

N_e - electron density

μ_B - Bohr magneton

The magnetic field in the foil is measured by changing the external magnetic field and measuring the voltage on a pickup coil wound around the foil:

$$B_{foil} = \Phi \cdot width \cdot thick, \quad \frac{d\Phi}{dt} = \varepsilon(t), \quad \Phi = \Phi_{foil} + \Phi_{air}$$

Φ_{air} is measured at a high field with the foil saturated.

Current results:

- Consistent results have been obtained with 3 samples of supermendur at $200\text{-}800\text{ Gs}$. Maximum polarization at 800 Gs is 8.3% , consistent with other measurements.
- Changing various parameters like frequency of the field variation etc, a maximum deviation of the result of $\pm 1.5\%$ was observed.
- The foil polarization at the target angles $20\text{-}30^\circ$ and a field of 250 Gs (used in Hall A) is 7.6%

Target Polarization

Additional systematic study:

- Measurements of the target thickness: relative thickness of the supermendur foil is uniform within 1%.
- Comparison of 13.7 μm supermendur and 5.0 μm iron targets: magnetic measurements and cross calibration with the beam are consistent within 1%.
Magnetization: Iron/Supermendur=0.929,
beam:0.933)

The relative systematic error: 2.%

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Dead Time

Discriminator: dead time measured $\tau \approx 20$ ns.

Test with LED: signals close to the signals from particles

Coincidence: $LG_{left} \times Ap_{left} \times LG_{right} \times Ap_{right}$

Møller measurements at HAPPEX: beam 3.35 GeV/c,
0.4 μ A, open collimator:

LG - 0.5 MHz, Ap - 1.0 MHz expected dead time $\approx 2\%$

μ A	Ap MHz	LED rate	Coinc.	$\mathcal{P}$
0.2	0.5	-1.3%		68.4 ± 0.2
0.4	1.0	-2.6%		68.2 ± 0.2
1.0	3.0	-6.7%		67.3 ± 0.2
2.0	5.5	-12.8%	-2.5%	65.3 ± 0.2 -4%

μ A	coll	Ap MHz	LED rate	Coinc.	$\mathcal{P}$
0.29	3.0cm	0.8	-2.2%		64.7 ± 0.2
1.10	3.0cm	3.4	-8.4%		64.69 ± 0.12
1.10	1.5cm	2.0	-5.4%		65.24 ± 0.12

The observed effect on polarization is below 1% at 0.4 μ A,
assigned error 2%

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Cross Calibration

Changing the RF phases of the lasers allowed to provide beam for each hall with all 3 lasers.

Polarimeter	Laser A	Laser B	Laser C
Mott	73%	74%	
Møller A	$69.2 \pm 0.15\%$	$68.2 \pm 0.15\%$	$-40.5 \pm 0.15\%$
Møller B	63.7%	$67.8 \pm 0.9\%$	$-38.1 \pm 1.0\%$
Møller C	$68.7 \pm 0.3\%$	$65.6 \pm 0.6\%$	$-39.8 \pm 0.2\%$

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Conclusion

- Stable operation
- Systematic errors estimated 3%
- More studies of the systematic errors are on the way:
 - target: another material
 - dead time
 - Cross-calibration with other polarimeters in JLAB
- New hardware
 - new power supplies for the quads
 - radiation hard detector
 - 2 new quads downstream of the Møller detector
(New constraints from Compton polarimeter forbid a common magnet settings for physics, Møller and Compton polarimeters)